

ROUSON DAS

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Summary

Cybersecurity student ranked top 15% globally on TryHackMe (57+ rooms, 3M+ users). Engineered 4 open-source security tools with CVE/NVD API integration, automated threat detection, and a 7-pattern IDS engine. Conducted vulnerability assessment and penetration testing across web, network, and forensics domains; published 3 peer-cited CTF writeups. Seeking an offensive security or VAPT internship to apply hands-on penetration testing skills in real-world engagements.

Education

B.Tech in Electronics & Communication Engineering

Aug 2023 – Jun 2027

Maulana Abul Kalam Azad University of Technology (MAKAUT) — Kolkata, India

- Relevant coursework: Network Security, Network Fundamentals, Digital Systems, Signals & Systems, VLSI Design, Control Systems.
- Participated in Smart India Hackathon and Nexus hackathons; preparing a Scopus-indexed cybersecurity research paper (target: Q3 2025).

Technical Skills

Penetration Testing	Vulnerability assessment, reconnaissance, enumeration, exploitation, privilege escalation, post-exploitation
Web App Security	OWASP Top 10, SQL injection, XSS, IDOR, CSRF, authentication bypass; Burp Suite interception & replay
Security Tools	Nmap, Burp Suite, Metasploit, ffuf, Gobuster, Wireshark, linPEAS, Netcat, Nikto, OWASP ZAP
Networking	TCP/IP, DNS, HTTP/S, SSH; Wireshark packet analysis; service fingerprinting; network forensics
Programming	Python (scripting, socket programming, automation), Bash shell scripting
Operating Systems	Kali Linux, Ubuntu, Windows; proficient across CLI and GUI environments
Version Control	Git, GitHub; 5+ documented public repositories
Leadership	Technical project ownership, peer mentoring, public technical presentation, open-source stewardship

Work Experience

Security Research Intern (Self-Directed)

Sep 2023 – Present

Open-Source Security Projects & CTF Research — Remote

- Achieved top 15% global ranking on TryHackMe (57+ rooms, 3M+ users) across web exploitation, privilege escalation, and network forensics.
- Identified and exploited 10+ OWASP Top 10 vulnerabilities (SQLi, XSS, IDOR, command injection) in structured lab environments; documented full attack chains and remediation steps for each.
- Intercepted and modified 100+ live HTTP requests using Burp Suite, uncovering authentication bypass and parameter tampering vulnerabilities across web application security labs.
- Enumerated 15+ target machines with Nmap and Gobuster; produced structured vulnerability reports mapping open ports and running services to matched CVEs for each target.
- Led a 3-member team during Chakravyuha 2.0 CTF, solving web challenges involving JavaScript credential extraction, traffic analysis, and steganographic image decoding.
- Published 3 peer-cited security writeups covering vulnerability assessment methodology, exploitation techniques, and remediation guidance.
- Evaluated 10+ community-submitted CTF solutions and penetration testing reports as volunteer peer reviewer in open-source security forums.

Projects

C2 Chat App — *Python, TCP, IDS, Remote Shell*

- Architected a multi-threaded TCP server (one thread per client + daemon console thread) with `threading.Lock`-guarded shared state across 10+ concurrent data structures (clients, admins, frozen, shadowed, watchlist, rate buckets, rshell/ping event maps).
- Built a 7-pattern IDS engine using compiled regex with severity tiers (CRITICAL → LOW), detecting destructive commands, reverse shells, download attempts, SQLi, XSS, and privilege escalation attempts in real time.
- Implemented token-bucket rate limiting (configurable burst/window via `server_config.ini`), persistent IP banning with temp-ban auto-expiry threads, and ghost-mute (shadow mode) where silenced users see their own messages but others don't.
- Engineered a bidirectional remote shell over the chat protocol using custom framing tags (`__RSHELL_REQ__` / `__RSHELL_RES__` / `__RSHELL_END__`) with `threading.Event`-based async result collection and configurable timeout.
- Delivered dual-format audit logging (plaintext and JSON), CSV export, hot-reloadable config (`/reload`), and a structured operator console with 30+ commands including `/inspect`, `/sniff`, `/ping`, `/tempban`, and `/rshell`.

Python Vulnerability Scanner — *NVD/CVE Integration*

- Engineered a CLI vulnerability scanner correlating discovered services against the NVD API, surfacing active CVEs with CVSS scores across 200+ service-version combinations.
- Reduced manual reconnaissance time by 60% in lab environments by automating TCP/UDP port scanning, banner grabbing, and structured JSON/CSV report generation.

Multi-Threaded Port Scanner — *Python*

- Designed a configurable scanner covering all 65,535 ports with thread-pool parallelism, cutting scan time by 75% vs. sequential baseline; supports CSV/JSON export for team reporting.

Certifications

NativeSOC Fundamentals — *NativeSOC; SOC operations, threat detection, and incident response* 2025

Advent of Cyber 2025 — *TryHackMe; web exploitation, forensics, networking, OSINT* 2025

Dark Web Monitoring Challenge — *HaxCamp by HaxSecurity; dark web threat intelligence, OSINT monitoring, and exposure analysis* Sep 2025